Module 4 – Functional Dependencies & Normalization

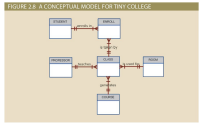
Learning Outcome (LO2): Apply techniques of database design including functional dependencies and normalization.

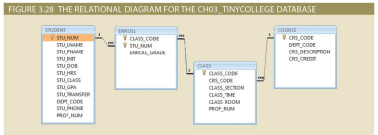
So far ...

Database Design → Gets Mapped → **Table Structures**

Model / Blueprint

ERD → **Relational Model**





How do we evaluate whether our table structures are good

What you will learn …

At the end of this class you will be able to:

- Explain **what** normalization is
- Explain **why** we normalize database tables
- Explain **the process** of normalization
- Discuss situations where **denormalization** might be required

How do we evaluate whether our table structures are good

Table structures are **good** when redundancy is controlled.

Because → **Redundancy causes anomalies**:

- **Update anomalies** (need to change many rows)
- **Insertion anomalies** (can't add data without unrelated info)
- **Deletion anomalies** (losing facts when deleting rows)

Examples of Anomalies

Update anomalies (e.g., in the STUDENT–COURSE table: schema STUDENT_COURSE(StudentID, CourseID, Instructor). If Instructor changes, many rows for that course must be updated)

Insertion anomalies (e.g., can't add a new course until at least one student enrolls)

Deletion anomalies (e.g., deleting the last student from a course may also remove the course information)

- **Bonus points** for producing a visual to show these tables and the redundant data in them. So we can see the anomalies.

The process of **finding** and then **eliminating** unnecessary/unwanted redundancies is called **Normalization**

Finding Redundancies

- To find redundancies, we need to **examine the relationships between attributes** of an entity.

- We are looking to find **Functional Dependencies (FDs)**

Functional Dependencies (FDs)

- Definition: Attribute set X → Y if **each value of X maps to exactly one Y.**
- Example:
  - PROJ_NUM → PROJ_NAME
  - EMP_NUM → EMP_NAME
- Notation: **X → Y**

Table name: DATA_ORG_1NF

PROJ_NUM	PROJ_NAME	EMP_NUM	
15	Evergreen	103	June
15	Evergreen	101	John
15	Evergreen	105	Alic
15	Evergreen	106	Willi
15	Evergreen	102	Davi
18	Amber Wave	114	Ann
18	Amber Wave	118	Jam
18	Amber Wave	104	Ann
18	Amber Wave	112	Darl
22	Rolling Tide	105	Alic
22	Rolling Tide	104	Ann

Keys and Dependencies

- **Superkey:** set of attributes uniquely identifying rows.
- **Candidate key:** minimal superkey.
- **Primary key:** chosen candidate key.
- Use FDs to validate keys & discover redundancy.

Our goal in analyzing FDs is to validate keys: to check that the chosen key really does uniquely identify tuples without including unnecessary attributes. **A good key should be minimal.**

Normalization Overview

- Goal: refine schema to reduce redundancy.
- Each normal form enforces stricter rules:
  - 1NF – no repeating groups.
  - 2NF – no partial dependencies.
  - 3NF – no transitive dependencies.
  - BCNF – every determinant is a superkey.

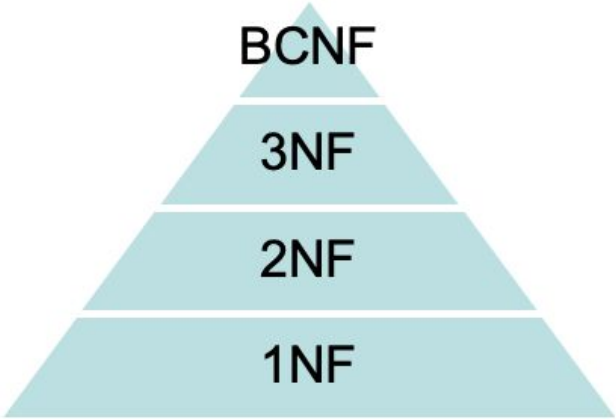


TABLE 6.1

A SAMPLE REPORT LAYOUT

PROJECT NUMBER	PROJECT NAME	EMPLOYEE NUMBER	EMPLOYEE NAME	JOB CLASS	CHARGE/ HOUR	HOURS BILLED	TOTAL CHARGE
15	Evergreen	103	June E. Arbough	Elec. Engineer	$ 84.50	23.8	$ 2,011.10
		101	John G. News	Database Designer	$105.00	19.4	$ 2,037.00
		105	Alice K. Johnson *	Database Designer	$105.00	35.7	$ 3,748.50
		106	William Smithfield	Programmer	$ 35.75	12.6	$ 450.45
		102	David H. Senior	Systems Analyst	$ 96.75	23.8	$ 2,302.65
				Subtotal			**$10,549.70**
18	Amber Wave	114	Annelise Jones	Applications Designer	$ 48.10	24.6	$ 1,183.26
		118	James J. Frommer	General Support	$ 18.36	45.3	$ 831.71
		104	Anne K. Ramoras *	Systems Analyst	$ 96.75	32.4	$ 3,134.70
		112	Darlene M. Smithson	DSS Analyst	$ 45.95	44.0	$ 2,021.80
				Subtotal			**$ 7,171.47**
22	Rolling Tide	105	Alice K. Johnson	Database Designer	$105.00	64.7	$ 6,793.50
		104	Anne K. Ramoras	Systems Analyst	$96.75	48.4	$ 4,682.70
		113	Delbert K. Joenbrood *	Applications Designer	$48.10	23.6	$ 1,135.16
		111	Geoff B. Wabash	Clerical Support	$26.87	22.0	$ 591.14
		106	William Smithfield	Programmer	$35.75	12.8	$ 457.60
				Subtotal			**$13,660.10**
25	Starflight	107	Maria D. Alonzo	Programmer	$ 35.75	24.6	$ 879.45
		115	Travis B. Bawangi	Systems Analyst	$ 96.75	45.8	$ 4,431.15
		101	John G. News *	Database Designer	$105.00	56.3	$ 5,911.50
		114	Annelise Jones	Applications Designer	$ 48.10	33.1	$ 1,592.11
		108	Ralph B. Washington	Systems Analyst	$ 96.75	23.6	$ 2,283.30
		118	James J. Frommer	General Support	$ 18.36	30.5	$ 559.98
		112	Darlene M. Smithson	DSS Analyst	$ 45.95	41.4	$ 1,902.33
				Subtotal			**$17,559.82**
				Total			**$48,941.09**

Note: * indicates the project leader.

Sample Report for a construction company that wants to bill its clients based on all the job done for their project.

A single report can be an **excellent starting point** to identify the base data necessary (to produce the report)

So the DB designer records the **base data** in a table structure

FIGURE 6.1 BASE DATA FOR A CONSTRUCTION COMPANY REPORT

Table name: RPT_FORMAT **Database name: Ch06_ConstructCo**

PROJ_NUM	PROJECT_NAME	EMP_NUMBER	EMP_NAME	JOB_CLASS	CHARGE_HOUR	HOURS_BILLED
15	Evergreen	103,101,105, 106, 102	June E. Arbough, John G. News, Alice K. Johnson *, William Smithfield, David H. Senior	Elec. Engineer, Database Designer, Database Designer, Programmer, System Analyst	85.5, 105., 105., 35.75, 98.75	23.8, 19.4, 35.7, 12.6, 23.8
18	Amber Wave	114, 118, 104, 112	Annelise Jones, James J. Frommer, Anne K. Ramoras *, Darlene M. Smithson	Applications Designer, General Support, Systems Analyst, DSS Analyst	48.1, 18.36, 96.75, 45.95	25.6, 45.3, 32.4, 45.
22	Rolling Tide	105, 104, 113, 111, 106	Alice K. Johnson, Anne K. Ramoras, Delbert K. Joenbrood *, Geoff B. Wabash, William Smithfield	DB Designer, Systems Analyst, Applications Designer, Clerical Support, Programmer	105., 96.75, 48.1, 26.87, 35.75	65.7, 48.4, 23.6, 22., 12.8
25	Star Light	107, 115, 101, 114, 108, 118, 112	Maria D. Alonzo, Travis B. Bawangi, John G. News *, Annelise Jones, Ralph B. Washington, James J. Frommer, Darlene M. Smithson	Programmer, Systems Analyst, Database Design, Applications Designer, Systems Analyst, General Support, DSS Analyst	35.75, 96.75, 105., 48.1, 96.75, 18.36, 45.95	25.6, 45.8, 56.3, 33.1, 23.6, 30.5, 41.4

This is **not a relational table**. Why?

TABLE 3.1

CHARACTERISTICS OF A RELATIONAL TABLE	
1	A table is perceived as a two-dimensional structure composed of rows and columns.
2	Each table row (**tuple**) represents a single entity occurrence within the entity set.
3	Each table column represents an attribute, and each column has a distinct name.
4	Each intersection of a row and column represents a single data value.
5	All values in a column must conform to the same data format.
6	Each column has a specific range of values known as the **attribute domain**.
7	The order of the rows and columns is immaterial to the DBMS.
8	Each table must have an attribute or combination of attributes that uniquely identifies each row.

There are **multiple values** in a single cells

First Normal Form (1NF)

- Each cell holds one value (atomic).
- Rows uniquely identified.

Table name: RPT_FORMAT — Database name: Ch06_ConstructCo

PROJ_NUM	PROJECT_NAME	EMP_NUMBER	EMP_NAME	JOB_CLASS	CHARGE_HOUR	HOURS_BILLED
15	Evergreen	103,101,105, 106, 102	June E. Arbough, John G. News, Alice K. Johnson *, William Smithfield, David H. Senior	Elec. Engineer, Database Designer, Database Designer, Programmer, System Analyst	85.5, 105., 105., 35.75, 98.75	23.8, 19.4, 35.7, 12.6, 23.8
18	Amber Wave	114, 118, 104, 112	Annelise Jones, James J. Frommer, Anne K. Ramoras *, Darlene M. Smithson	Applications Designer, General Support, Systems Analyst, DSS Analyst	48.1, 18.36, 96.75, 45.95	25.6, 45.3, 32.4, 45.
22	Rolling Tide	105, 104, 113, 111, 106	Alice K. Johnson, Anne K. Ramoras, Delbert K. Joenbrood *, Geoff B. Wabash, William Smithfield	DB Designer, Systems Analyst, Applications Designer, Clerical Support, Programmer	105., 96.75, 48.1, 26.87, 35.75	65.7, 48.4, 23.6, 22., 12.8
25	Star Light	107, 115, 101, 114, 108, 118, 112	Maria D. Alonzo, Travis B. Bawangi, John G. News *, Annelise Jones, Ralph B. Washington, James J. Frommer, Darlene M. Smithson	Programmer, Systems Analyst, Database Design, Applications Designer, Systems Analyst, General Support, DSS Analyst	35.75, 96.75, 105., 48.1, 96.75, 18.36, 45.95	25.6, 45.8, 56.3, 33.1, 23.6, 30.5, 41.4

Before (with repeating groups of attributes)

PK

Table name: DATA_ORG_1NF — Database name: Ch06_ConstructCo

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
15	Evergreen	105	Alice K. Johnson *	Database Designer	105.00	35.7
15	Evergreen	106	William Smithfield	Programmer	35.75	12.6
15	Evergreen	102	David H. Senior	Systems Analyst	96.75	23.8
18	Amber Wave	114	Annelise Jones	Applications Designer	48.10	24.6
18	Amber Wave	118	James J. Frommer	General Support	18.36	45.3
18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	96.75	32.4
18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	45.95	44.0
22	Rolling Tide	105	Alice K. Johnson	Database Designer	105.00	64.7
22	Rolling Tide	104	Anne K. Ramoras	Systems Analyst	96.75	48.4
22	Rolling Tide	113	Delbert K. Joenbrood *	Applications Designer	48.10	23.6
22	Rolling Tide	111	Geoff B. Wabash	Clerical Support	26.87	22.0
22	Rolling Tide	106	William Smithfield	Programmer	35.75	12.8
25	Starflight	107	Maria D. Alonzo	Programmer	35.75	24.6
25	Starflight	115	Travis B. Bawangi	Systems Analyst	96.75	45.8
25	Starflight	101	John G. News *	Database Designer	105.00	56.3
25	Starflight	114	Annelise Jones	Applications Designer	48.10	33.1
25	Starflight	108	Ralph B. Washington	Systems Analyst	96.75	23.6
25	Starflight	118	James J. Frommer	General Support	18.36	30.5
25	Starflight	112	Darlene M. Smithson	DSS Analyst	45.95	41.4

After (removing repeating groups)

Eliminating Repeating Groups

To normalize a table, we first put it in a **proper table format** so that each row and column -- each cell -- represents a **single value**.

FIGURE 6.1 BASE DATA FOR A CONSTRUCTION COMPANY REPORT

Table name: RPT_FORMAT **Database name: Ch06_ConstructCo**

PROJ_NUM	PROJECT_NAME	EMP_NUMBER	EMP_NAME	JOB_CLASS	CHARGE_HOUR	HOURS_BILLED
15	Evergreen	103,101,105, 106, 102	June E. Arbough, John G. News, Alice K. Johnson *, William Smithfield, David H. Senior	Elec. Engineer, Database Designer, Database Designer, Programmer, System Analyst	85.5, 105., 105., 35.75, 98.75	23.8, 19.4, 35.7, 12.6, 23.8
18	Amber Wave	114, 118, 104, 112	Annelise Jones, James J. Frommer, Anne K. Ramoras *, Darlene M. Smithson	Applications Designer, General Support, Systems Analyst, DSS Analyst	48.1, 18.36, 96.75, 45.95	25.6, 45.3, 32.4, 45.
22	Rolling Tide	105, 104, 113, 111, 106	Alice K. Johnson, Anne K. Ramoras, Delbert K. Joenbrood *, Geoff B. Wabash, William Smithfield	DB Designer, Systems Analyst, Applications Designer, Clerical Support, Programmer	105., 96.75, 48.1, 26.87, 35.75	65.7, 48.4, 23.6, 22., 12.8
25	Star Light	107, 115, 101, 114, 108, 118, 112	Maria D. Alonzo, Travis B. Bawangi, John G. News *, Annelise Jones, Ralph B. Washington, James J. Frommer, Darlene M. Smithson	Programmer, Systems Analyst, Database Design, Applications Designer, Systems Analyst, General Support, DSS Analyst	35.75, 96.75, 105., 48.1, 96.75, 18.36, 45.95	25.6, 45.8, 56.3, 33.1, 23.6, 30.5, 41.4

Before: Each row represented information about a single project. Including all people working on it and the work they've done → So the **PK was PROJ_NUM**

FIGURE 6.2 A TABLE IN FIRST NORMAL FORM

Table name: DATA_ORG_1NF **Database name: Ch06_ConstructCo**

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
15	Evergreen	105	Alice K. Johnson *	Database Designer	105.00	35.7
15	Evergreen	106	William Smithfield	Programmer	35.75	12.6
15	Evergreen	102	David H. Senior	Systems Analyst	96.75	23.8
18	Amber Wave	114	Annelise Jones	Applications Designer	48.10	24.6
18	Amber Wave	118	James J. Frommer	General Support	18.36	45.3
18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	96.75	32.4
18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	45.95	44.0
22	Rolling Tide	105	Alice K. Johnson	Database Designer	105.00	64.7

Each row represents the work of a particular employee in a specific project, so both PROJ_NUM and EMP_NUM identify the row →

PK is (PROJ_NUM + EMP_NUM)

The Normalization Process

Notice that so far we have:

1. **eliminated repeating groups** of attributes so that each row and column contains only one value.
2. identified the **new Primary Key**
3. ...

3. Identify **Functional Dependencies**

Partial Dependency

- Occurs when a non-key attribute depends only on **part** of a composite key (and not the whole key)

PK

Table name: DATA_ORG_1NF Database name: Ch06_ConstructCo

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
15	Evergreen	105	Alice K. Johnson *	Database Designer	105.00	35.7
15	Evergreen	106	William Smithfield	Programmer	35.75	12.6
15	Evergreen	102	David H. Senior	Systems Analyst	96.75	23.8
18	Amber Wave	114	Annelise Jones	Applications Designer	48.10	24.6
18	Amber Wave	118	James J. Frommer	General Support	18.36	45.3
18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	96.75	32.4
18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	45.95	44.0
22	Rolling Tide	105	Alice K. Johnson	Database Designer	105.00	64.7
22	Rolling Tide	104	Anne K. Ramoras	Systems Analyst	96.75	48.4
22	Rolling Tide	113	Delbert K. Joenbrood *	Applications Designer	48.10	23.6
22	Rolling Tide	111	Geoff B. Wabash	Clerical Support	26.87	22.0
22	Rolling Tide	106	William Smithfield	Programmer	35.75	12.8
25	Starflight	107	Maria D. Alonzo	Programmer	35.75	24.6
25	Starflight	115	Travis B. Bawangi	Systems Analyst	96.75	45.8
25	Starflight	101	John G. News *	Database Designer	105.00	56.3
25	Starflight	114	Annelise Jones	Applications Designer	48.10	33.1
25	Starflight	108	Ralph B. Washington	Systems Analyst	96.75	23.6
25	Starflight	118	James J. Frommer	General Support	18.36	30.5
25	Starflight	112	Darlene M. Smithson	DSS Analyst	45.95	41.4

- Can you detect any partial dependencies in the 1NF table we have so far?

FIGURE 6.2 A TABLE IN FIRST N

Table name: DATA_ORG_1NF

PROJ_NUM	PROJ_NAME	EMP_NUM	
15	Evergreen	103	June
15	Evergreen	101	John
15	Evergreen	105	Alic
15	Evergreen	106	Willi
15	Evergreen	102	Davi
18	Amber Wave	114	Ann
18	Amber Wave	118	Jam
18	Amber Wave	104	Ann
18	Amber Wave	112	Darl
22	Rolling Tide	105	Alic
22	Rolling Tide	104	Ann
22	Rolling Tide	113	Delb
22	Rolling Tide	111	Geo
22	Rolling Tide	106	Willi
25	Starflight	107	Mari
25	Starflight	115	Trav
25	Starflight	101	John
25	Starflight	114	Ann
25	Starflight	108	Ralp
25	Starflight	118	Jam
25	Starflight	112	Darl

- Notice how PROJ_NAME depends only on PROJ_NUM.
- Knowing a project's number, we can determine the project's name because only one name is associated with any given project number.
- I.e. we do not need to know the the entire key (PROJ_NUM, EMP_NUM) to determine PROJ_NAME. Which tells us the key in this table is **not a good key as it is not minimal.**

PROJ_NUM→ PROJ_NAME

FIGURE 6.2 A TABLE IN FIRST NORMAL FORM

Table name: DATA_ORG_1NF **Database name: Ch06_ConstructCo**

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
15	Evergreen	105	Alice K. Johnson *	Database Designer	105.00	35.7
15	Evergreen	106	William Smithfield	Programmer	35.75	12.6
15	Evergreen	102	David H. Senior	Systems Analyst	96.75	23.8
18	Amber Wave	114	Annelise Jones	Applications Designer	48.10	24.6
18	Amber Wave	118	James J. Frommer	General Support	18.36	45.3
18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	96.75	32.4
18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	45.95	44.0
22	Rolling Tide	105	Alice K. Johnson	Database Designer	105.00	64.7
22	Rolling Tide	104	Anne K. Ramoras	Systems Analyst	96.75	48.4
22	Rolling Tide	113	Delbert K. Joenbrood *	Applications Designer	48.10	23.6

Similarly, given an employee's number alone, we would know their name, job class and charge per hour:

EMP_NUM → (EMP_NAME, JOB_CLASS, CHG_HOUR)

Partial Dependency

- Occurs when a non-key attribute depends only on **part** of a composite key (and not the whole key)

PK

Table name: DATA_ORG_1NF Database name: Ch06_ConstructCo

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
15	Evergreen	105	Alice K. Johnson *	Database Designer	105.00	35.7
15	Evergreen	106	William Smithfield	Programmer	35.75	12.6
15	Evergreen	102	David H. Senior	Systems Analyst	96.75	23.8
18	Amber Wave	114	Annelise Jones	Applications Designer	48.10	24.6
18	Amber Wave	118	James J. Frommer	General Support	18.36	45.3
18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	96.75	32.4
18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	45.95	44.0
22	Rolling Tide	105	Alice K. Johnson	Database Designer	105.00	64.7
22	Rolling Tide	104	Anne K. Ramoras	Systems Analyst	96.75	48.4
22	Rolling Tide	113	Delbert K. Joenbrood *	Applications Designer	48.10	23.6
22	Rolling Tide	111	Geoff B. Wabash	Clerical Support	26.87	22.0
22	Rolling Tide	106	William Smithfield	Programmer	35.75	12.8
25	Starflight	107	Maria D. Alonzo	Programmer	35.75	24.6
25	Starflight	115	Travis B. Bawangi	Systems Analyst	96.75	45.8
25	Starflight	101	John G. News *	Database Designer	105.00	56.3
25	Starflight	114	Annelise Jones	Applications Designer	48.10	33.1
25	Starflight	108	Ralph B. Washington	Systems Analyst	96.75	23.6
25	Starflight	118	James J. Frommer	General Support	18.36	30.5
25	Starflight	112	Darlene M. Smithson	DSS Analyst	45.95	41.4

Two partial dependencies in this table:

PROJ_NUM → PROJ_NAME

EMP_NUM →
EMP_NAME, JOB_CLASS, CHG_HOUR

FIGURE 6.2 A TABLE IN FIRST NORMAL FORM

Table name: DATA_ORG_1NF **Database name: Ch06_ConstructCo**

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
15	Evergreen	105	Alice K. Johnson *	Database Designer	105.00	35.7
15	Evergreen	106	William Smithfield	Programmer	35.75	12.6
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18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	96.75	32.4
18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	45.95	44.0
22	Rolling Tide	105	Alice K. Johnson	Database Designer	105.00	64.7

Consider **anomalies that could** happen if we want to:

1. update the JOB_CLASS → should update many entries
2. insert a new employee → should be associated with a project / need for a phantom entity
3. delete an employee → project info might get deleted!

Also nothing prevents the user from typing slightly different versions of the same data

A Table in 1NF (i.e. any relational table) can still be subject to data redundancies, if there are any partial dependencies.

Second Normal Form (2NF)

2NF = 1NF + no **partial dependencies.**

To remove partial dependencies, we:

1. Create a separate table for each partial dependency
2. Reassign dependent attributes

$$\underbrace{X}_{\text{Determinant}} \rightarrow \underbrace{Y}_{\text{Dependent}}$$

1. Create New Tables

The determinants of the PDs each become the PK of a new table:

PROJECT(**PROJ_NUM**)

EMPLOYEE(**EMP_NUM**)

New Tables

Also, the **original table** remains in the DB:

ASSIGNMENT(**PROJ_NUM, EMP_NUM**)

2. Re-assign Dependent Attributes

The **dependent attributes** are moved over and removed from the original table

PROJECT (**PRJ_NUM**, PROJ_NAME)

EMPLOYEE (**EMP_NUM**, EMP_NAME, JOB_CLASS, CHG_HOUR)

ASSIGNMENT (**PROJ_NUM**, **EMP_NUM**, ASSIGN_HOUR)

Any attribute not partially dependent remains in the **original table**

FIGURE 6.4 SECOND NORMAL FORM (2NF) CONVERSION RESULTS

Table name: PROJECT

PROJ_NUM | PROJ_NAME

PROJECT (PROJ_NUM, PROJ_NAME)

Table name: EMPLOYEE

EMP_NUM | EMP_NAME | JOB_CLASS | CHG_HOUR

Transitive dependency

EMPLOYEE (EMP_NUM, EMP_NAME, JOB_CLASS, CHG_HOUR)

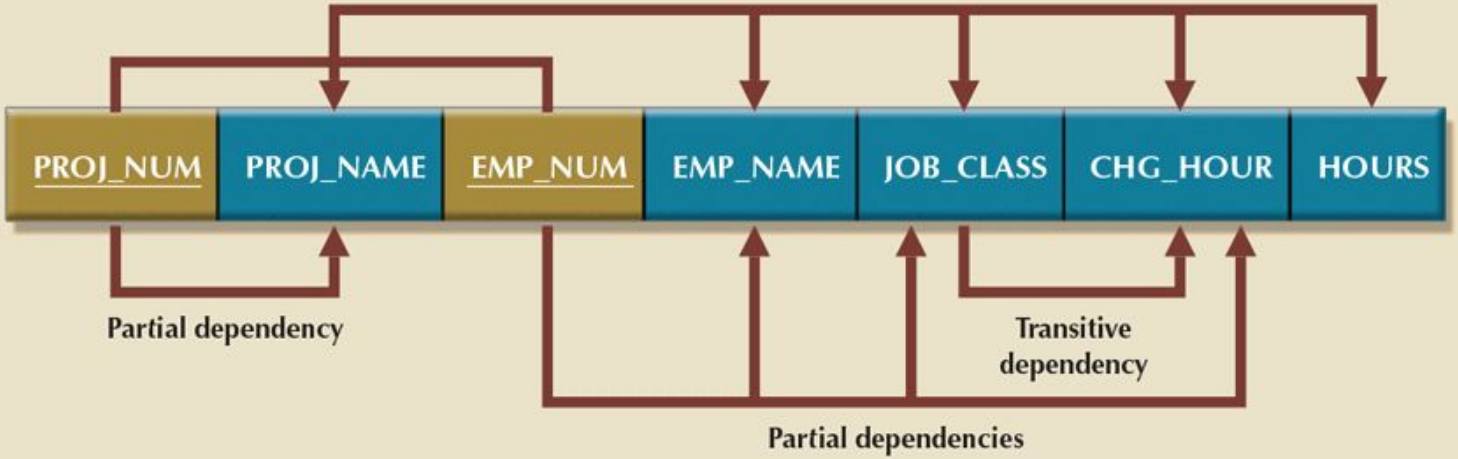
TRANSITIVE DEPENDENCY
(JOB_CLASS → CHG_HOUR)

Table name: ASSIGNMENT

PROJ_NUM | EMP_NUM | ASSIGN_HOURS

ASSIGNMENT (PROJ_NUM, EMP_NUM, ASSIGN_HOURS)

FIGURE 6.3 FIRST NORMAL FORM (1NF) DEPENDENCY DIAGRAM



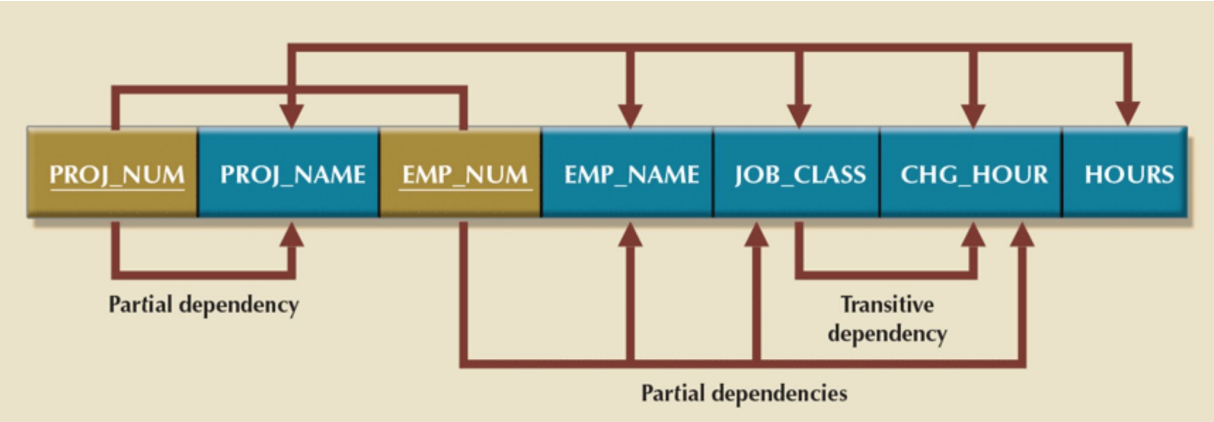
1NF (PROJ_NUM, EMP_NUM, PROJ_NAME, EMP_NAME, JOB_CLASS, CHG_HOURS, HOURS)

PARTIAL DEPENDENCIES:
(PROJ_NUM → PROJ_NAME)
(EMP_NUM → EMP_NAME, JOB_CLASS, CHG_HOUR)

TRANSITIVE DEPENDENCY:
(JOB_CLASS → CHG_HOUR)

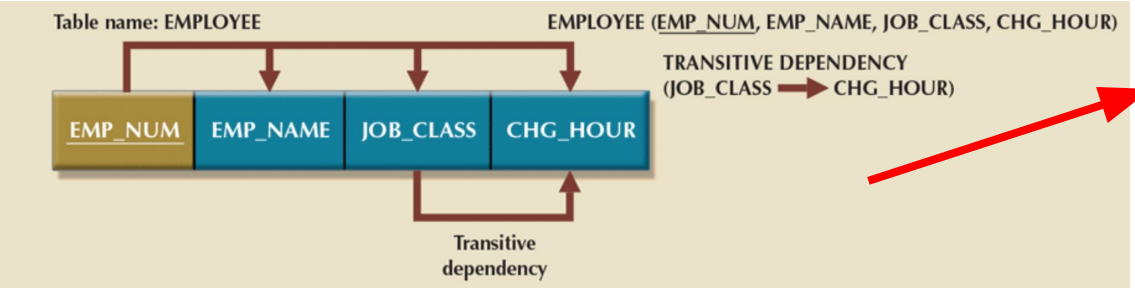
Transitive Dependency

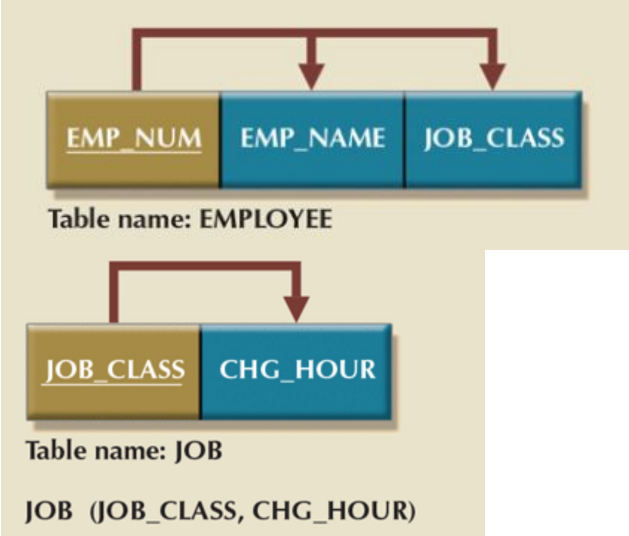
- Occurs when we have attributes in a table that only indirectly depend on the key. In fact they depend on an attribute which is not a candidate key in this table.
- Example: EMP_NUM → JOB_CLASS→ CHG_HOUR



Eliminating transitive dependencies

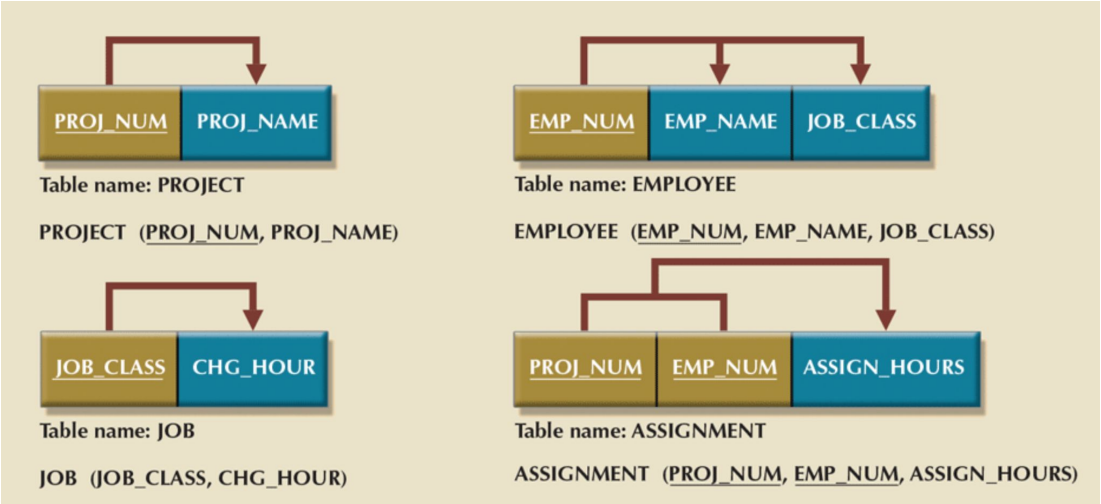
- Create a new table for each dependency.
- Move the dependant attribute(s) to the new table.
- Keep a copy of the determinant in the original.





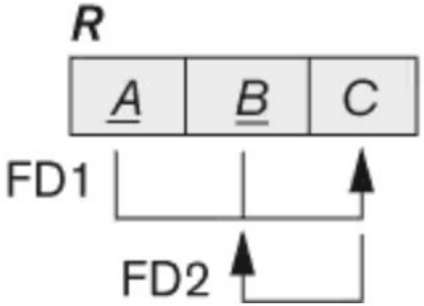
Third Normal Form (3NF)

3 NF = 2NF + no **transitive dependencies:**



Boyce-Codd Normal Form (BCNF)

- Refinement of 3NF.
- **Difference from 3NF:** 3NF allows certain exceptions (if the dependent attribute is part of a candidate key), but BCNF removes all exceptions.
- **Rule:** For every FD X → Y, X must be a superkey.
- **Practical note:** In most cases, if a schema is in 3NF it is also in BCNF. The only exception is when there exists an FD X → A where X is not a superkey and A is a prime attribute. In such cases the schema is in 3NF but not BCNF.



Decomposition Principles

When decomposing, ensure:

1. **Lossless-join:** join of decomposed tables = original table.
   a. Rule: shared attribute must be a key in one table.
2. **Dependency preservation:** all FDs must still hold.

The Normalization Process

1. **eliminate repeating groups** of attributes so that each row and column contains only one value.
2. identify the **new Primary Key**
3. Identify all **dependencies**
   a. Eliminating partial dependencies → 2NF
   b. Eliminating transitive dependencies → 3NF
4. Check BCNF
5. Verify lossless-joins + dependency preservation

Some Notes on Normalization

- Two ways to apply normalization:
  1. Examine initial ERD to see if the design can be improved by Normalization
  2. More often, the DB designer is asked to **modify existing data structures** (files, spreadsheets, or existing DB structures)
- Optimal implementation → requires at least 3NF
- **Normalization purity** is often difficult to sustain in the modern database environment
- Sometimes redundancy is kept intentionally:
  - For performance.
  - For reporting/analytics.
- Trade-off: efficiency vs consistency.
-

Summary

- FDs = discover hidden redundancy.
- Normalization = remove anomalies stepwise.
- 1NF → atomic values.
- 2NF → no partial dependencies.
- 3NF → no transitive dependencies.
- BCNF → every FD has a superkey.
- Always check: lossless-join + dependency preservation.